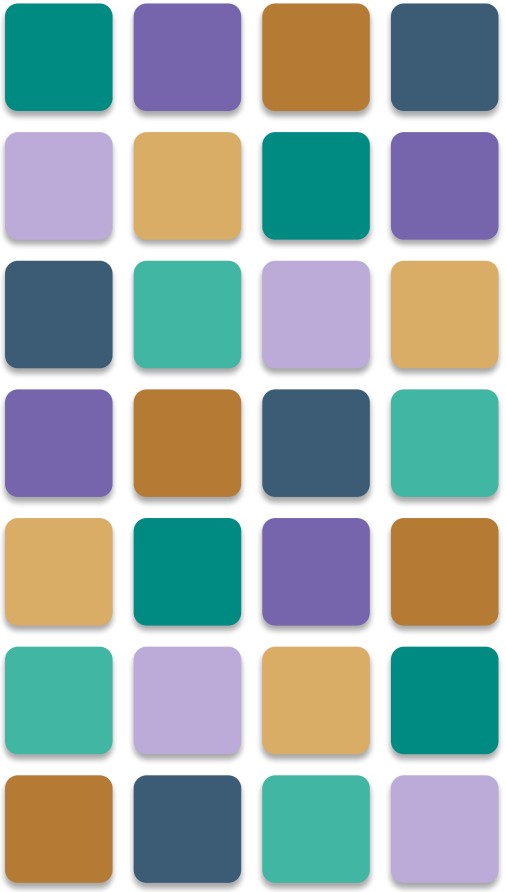
PHD

Supervised Semantic Segmentation Baselines for Remote Sensing

Model Selection, Evaluation and Predictive Reliability



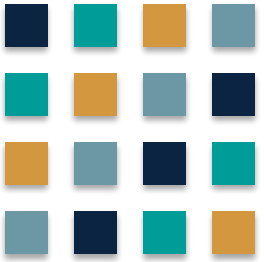
Wilgo Nunes

Institute of Systems and Robotics · University of Coimbra

october 2026

01

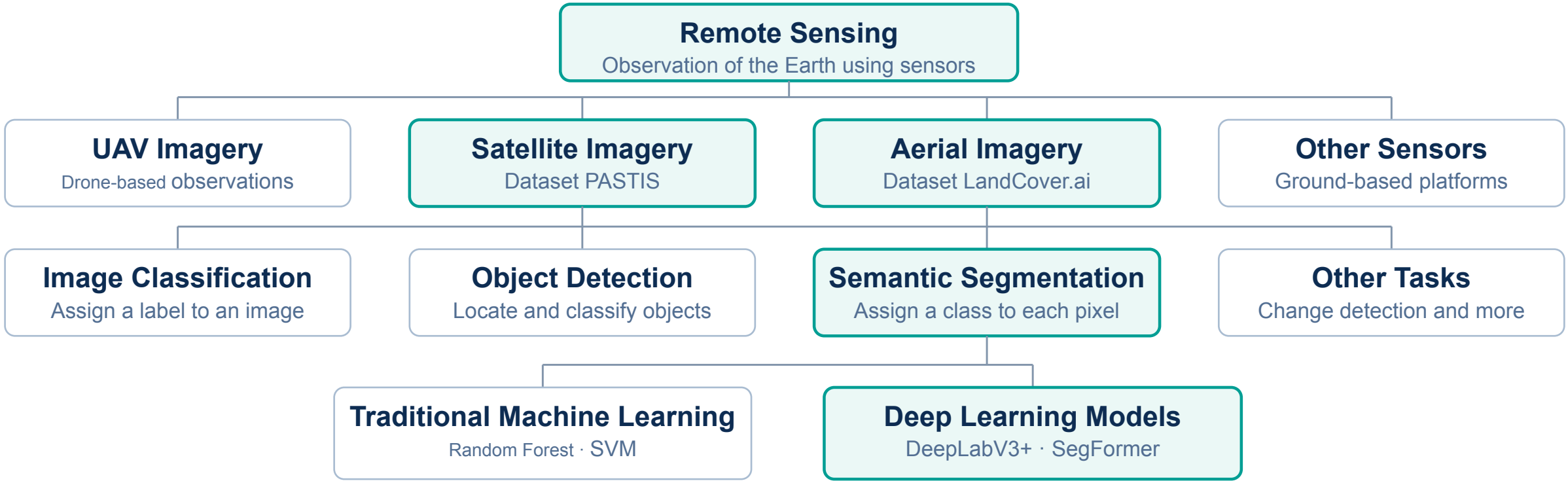
Research Context



Scientific context

Positioning within Remote Sensing

From remote sensing imagery to reliable semantic segmentation

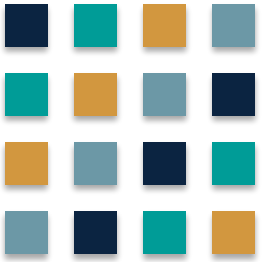


Establishing reproducible supervised baselines for subsequent calibration and knowledge distillation studies

Scope: satellite and aerial imagery, semantic segmentation

02

Datasets & Baseline Models



benchmarks · supervised segmentation architectures

Datasets & Baseline Models

Two remote sensing benchmarks · Two supervised segmentation architectures

DATASETS 02

PASTIS

Sentinel-2 · Crop segmentation · France

Garnot & Landrieu, ICCV 2021

LandCover.ai

Aerial RGB · Land-cover segmentation · Poland

Boguszewski et al., CVPRW 2021

BASELINES 02

DeepLab V3+ | CNN

Multi-scale context and boundary refinement

Chen et al., ECCV 2018

SegFormer-B1 | Transformer

Hierarchical features and lightweight decoder

Xie et al., NeurIPS 2021

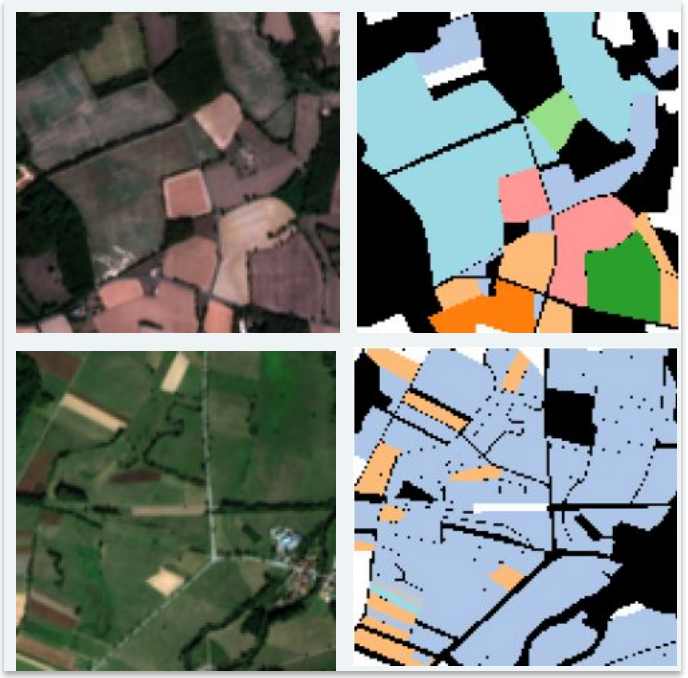
Next: dataset characteristics and the scientific rationale for each baseline architecture

Dataset PASTIS | Satellite crop segmentation

ICCV 2021 • Sentinel-2 • France

Sensor / platform	Sentinel-2 MSI (optical)
Spatial resolution	10 m analysis grid; native MSI bands: 10/20/60 m
Spectral resolution	10 input bands (VIS, red-edge, NIR, SWIR); band widths
Patch size	128 × 128 px • 1.28 × 1.28 km
Dataset size	2,433 patches • 38–61 acquisitions / patch
Classes / labels	18 crop types + background; semantic and instance
Region / period	4 Sentinel-2 tiles, France • Sep 2018–Nov 2019
Evaluation	Official 5-fold split
Our intended use	Single-date subset; selection rule fixed before training

RGB Image | Ground Truth



Why: Sentinel-2, public masks and official folds. Single-date selection is our adaptation; temporal baselines are not directly comparable.

Sources: Garnot & Landrieu, ICCV 2021; github.com/VSainteuf/pastis-benchmark

Dataset LandCover.ai | Aerial Land-Cover Segmentation

CVPR Workshops 2021 · Aerial orthophotos · Poland

Sensor / platform	Aerial orthophotography
Spatial resolution	0.25 / 0.50 m per pixel
Spectral resolution	RGB (3 bands)
Patch size	512 × 512 pixels
Dataset size	41 orthophotos · ~3 billion pixels
Classes / labels	Background + 4 land-cover classes
Region / period	Poland · 216 km²
Evaluation	Geographically grouped holdout
Our intended use	Supervised semantic segmentation

RGB Image | Ground Truth



Why: High-resolution RGB imagery, dense labels, and a geographically separated evaluation protocol.

Source: Boguszewski et al., CVPRW 2021 · landcover.ai.linuxpolska.com

Model Selection | Supervised Baselines

Two complementary architectures evaluated on both datasets

DeepLab V3+ | CNN

- Atrous spatial pyramid pooling: multi-scale context
- Decoder refines spatial details and boundaries
- Established supervised reference; used in LandCover.ai dataset comparisons

SegFormer-B1 | Transformer

- Hierarchical MiT encoder: multi-scale features
- Lightweight MLP decoder; moderate model size
- Modern, reproducible alternative to a CNN

Why not directly use the models from the dataset papers?

PASTIS | ICCV 2021

Original: U-TAE + PaPs; leaderboard also TSViT, temporal U-Nets.

Not selected: designed for image time series / panoptic tasks; our protocol uses single-date semantic segmentation.

LandCover.ai | CVPRW 2021

Original: land-cover mapping with CNN-based segmentation benchmarks.

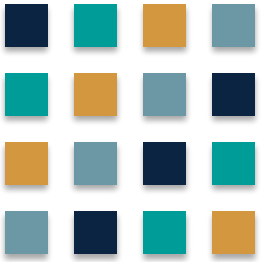
Our task: geographically separated evaluation with a common CNN–Transformer protocol.

Controlled comparison: consistent architectures, training objectives and evaluation metrics across both datasets.

Sources: Chen et al., ECCV 2018 · Xie et al., NeurIPS 2021 · Garnot & Landrieu, ICCV 2021 · Boguszewski et al., CVPRW 2021

03

Datasets Preprocessing



benchmarks datasets · supervised segmentation architectures

Dataset Splits & Evaluation Protocols

Two remote sensing benchmarks · Distinct evaluation protocols

PASTIS

Official five-fold split

PROTOCOL

Five official patch folds; rotating test fold

INDEPENDENCE

Disjoint patches, but spatial correlation remains

INTERPRETATION

Official-fold performance, not independent-region generalization

LandCover.ai

Custom geographically grouped split

PROTOCOL

25 train · 5 validation · 5 calibration · 6 test scenes

INDEPENDENCE

Geographic groups kept intact; minimum 8.11 km separation

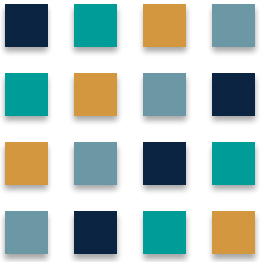
INTERPRETATION

Custom spatial holdout; not directly comparable to original random-chip results

Key distinction: PASTIS uses official folds; LandCover.ai prioritizes geographic separation.

04

Model Training & Selection



Supervised training · Hyperparameter comparison · Checkpoint selection · Held-out evaluation

Supervised Baseline | Experimental Design

Two architectures · Two datasets · Controlled model selection

FACTORIAL COMPARISON

2 architectures × 2 initializations × 2 LR schedules

DeepLabV3+ / SegFormer-B1 From scratch / Fine-tuning Fixed LR / LR scheduling

01 TRAIN & VALIDATE

Train each configuration; use Early Stopping and checkpoint-selection mIoU.

02 SELECT & REUSE

Choose the best checkpoint per architecture; no additional retraining.

03 EVALUATE

Evaluate selected checkpoints on held-out data; never select using test results.

PASTIS Official 5-fold test (patch-disjoint, spatially correlated) | LandCover.ai Geographic holdout

Validation = checkpoint selection • Calibration partitions reserved, not used at this stage • No soft labels or distillation
Fine-tuning: ImageNet-pretrained encoder, all parameters trainable. LR scheduling: PolynomialLR.

EXPERIMENTAL RESULTS

Model Selection | PASTIS & LandCover.ai

Checkpoint-selection validation mIoU (%) · Same 2 × 2 design

PASTIS · Run 01

CONFIGURATION	DeepLab	SegFormer
Scratch · Fixed LR	**25.40**	20.13
Scratch · Scheduled LR	24.83	20.34
Fine-tune · Fixed LR	20.03	**25.92**
Fine-tune · Scheduled LR	19.60	25.49

LandCover.ai

CONFIGURATION	DeepLab	SegFormer
Scratch · Fixed LR	57.64	55.55
Scratch · Scheduled LR	56.36	53.79
Fine-tune · Fixed LR	**69.67**	**68.28**
Fine-tune · Scheduled LR	68.64	67.42

PASTIS: architecture-dependent initialization · LandCover.ai: fine-tuning selected for both

Fine-tuning: ImageNet-pretrained trainable encoder · Scheduled LR: PolynomialLR · Fixed LR: no scheduler

Selection validation only; not held-out test performance. Single-seed factorial comparisons.

EXPERIMENTAL RESULTS

Supervised Baselines | Held-Out Test Results

Test mIoU, pixel accuracy and macro F1 · Different evaluation protocols

PASTIS · Official five-fold test

Mean ± sample SD

DeepLabV3+ · ResNet-50

From scratch · Fixed LR

mIoU 27.46%

Accuracy 68.85% · Macro F1 38.00%

SegFormer-B1

Fine-tuning · Fixed LR

mIoU 28.94%

Accuracy 70.46% · Macro F1 39.93%

LandCover.ai · Geographic holdout

6 scenes · Pixel-pooled

DeepLabV3+ · ResNet-50

Fine-tuning · Fixed LR

mIoU 73.57%

Accuracy 93.09% · Macro F1 83.89%

SegFormer-B1

Fine-tuning · Fixed LR

mIoU 73.86%

Accuracy 92.00% · Macro F1 84.25%

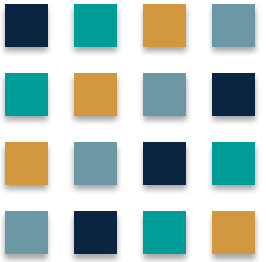
PASTIS: patch-disjoint, spatially correlated folds. LandCover.ai: geographically separated test scenes.

Descriptive comparisons only · No test-based checkpoint selection · No additional training

05

Baseline Reliability Assessment

Frozen supervised models · Uncalibrated predictions



Segmentation quality • Error detection • Selective prediction

UNCALIBRATED REFERENCE

Frozen Baseline

Segmentation & predictive reliability | Held-out test

METRIC	PASTIS		LANDCOVER.AI	
	DeepLab V3+	SegFormer-B1	DeepLab V3+	SegFormer-B1
mIoU ↑	0.275	0.289	0.736	0.739
Accuracy ↑	0.689	0.705	0.931	0.920
Macro F1 ↑	0.380	0.399	0.839	0.843
ECE ↓	0.021	0.095	0.028	0.029
AdaECE ↓	0.021	0.095	0.028	0.029
NLL ↓	0.974	1.037	0.218	0.245
Brier ↓	0.438	0.435	0.108	0.124
AUROC ↑	0.761	0.764	0.888	0.883
FPR95 ↓	0.664	0.681	0.429	0.380
AURC ↓	0.147	0.139	0.012	0.015
E-AURC ↓	0.092	0.091	0.010	0.012

PASTIS: mean across 5 official folds • LandCover.ai: pixel-pooled across 6 geographic test scenes

Fold SDs in full report • LandCover.ai ranking metrics approximated (262,144 bins) • AUSE omitted (equivalent to E-AURC)

Conclusions & Next Steps

Frozen Baseline v1.0 | PASTIS & LandCover.ai

01

Baseline established

Two datasets and two architectures evaluated with frozen checkpoints and documented test protocols.

02

Key observation

Higher segmentation performance does not necessarily imply better probability calibration.

03

Next research phase

Benchmark calibration methods, select candidates using development data, then evaluate their effect on knowledge distillation.

From reliable baselines to calibrated teacher supervision

PASTIS: five official folds • LandCover.ai: six geographically held-out test scenes • No post-hoc calibration or KD in v1.0